HCMA-Net: Hierarchical Cross-Modality Aggregation Network for Multimodal Remote Sensing Image Classification

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***Abstract*—While multimodal remote sensing images provide complementary information in different imaging ways, effectively aggregating and jointly learning from these heterogeneous features is nontrivial, primarily due to the inherent modality gap that hinders semantic alignment and feature interoperability. To overcome these issues, we propose a hierarchical cross-modality aggregation network (HCMA-Net). It introduces two novel components: the hierarchical feature aggregation and enhancement module (HFAE-Module) and the cross-modality interactive feature extraction module (CMIFE-Module). The HFAE-Module tackles the modality gap and enables cross-scale interaction through its hierarchical cross-modality feature aggregation (HCMFA) mechanism, which incorporates cross-spectral and spatial aggregation nonlocal attention (CSNLA and SANLA) layers to align features and aggregate contextual information across spectral and spatial dimensions. The CMIFE-Module addresses the optimization conflict by leveraging a dual-attention design; it uses self-attention to reinforce intramodal coherence and cross-attention to dynamically extract and fuse complementary intermodal features, thereby maximizing complementarity while avoiding the dilution of discriminative features and preventing negative transfer. Experiments on four real-world datasets (Hohhot, Nanjing, Xi'an, and Houston2013) demonstrate that HCMA-Net consistently achieves outstanding classification results. The code is available at: https://github.com/sun740936222/HCMA-Net**

***Index Terms*—Cross-modal feature alignment, image classification, multimodal, remote sensing.**

I. Introduction

REMOTE sensing technology has become a pivotal tool in diverse fields, including geographic information systems, environmental monitoring, and precision agriculture [1], [2], [3]. Advancements in sensor technology have led to remote sensing data with diverse resolutions—spatial, spectral, and temporal [4], [5]. Multimodal fusion of panchromatic (PAN) and multispectral (MS) images is particularly critical, as PAN provides high-resolution spatial details while MS offers rich spectral information. However, effectively integrating these heterogeneous modalities remains challenging due to inherent disparities in their feature representations and imaging characteristics.

In recent years, with the rapid development of deep learning technology [6], [7], [8], [9], the classification of fusion methods for the analysis of multimodal remote sensing images can be broadly categorized into two distinct approaches [10]: early fusion and late fusion, based on the multibranch architecture employed. For early fusion-based methods, information obtained from different sensors is fused at a relatively shallow position of the network structure. Ghamisi et al. [11] extracted the extinction profiles from Hyperspectral and LiDAR data and fused them on the deep feature space generated by deep convolutional neural networks (CNNs). Rasti et al. [12] applied extinction profiles and total variation component analysis to improve feature fusion. However, early fusion methods often oversimplify intermodal interactions through mechanisms like crosstalk and weighted summation, neglecting complex relationships.

Late fusion based on multibranch structure methods has demonstrated significant success in multimodal remote sensing image fusion and classification tasks [13], [14], [15], [16]. Xu et al. [17] initially introduced a dual-branch CNN network with the objective of enhancing classification results by fusing multisource remote sensing data. Hang et al. [18] identified that the work of Xu et al. [17] lacked effective integration of multisource features. Consequently, they proposed an innovative dual-branch coupled CNN architecture. It aimed to better capture and utilize spectral and spatial characteristics and elevation data, enhancing the overall feature learning process. Tu et al. [19] proposed a network combining global and local features for fusion (NCGLF^2) that integrated global and local features (GLFs) extracted from multisource remote sensing data. Zhao et al. [20] proposed a global collaborative fusion framework that leveraged the comprehensive contextual information embedded throughout an image to enhance multimodal data integration. Fang et al. [21] used a three-branch network architecture to maximize the utilization of ultrasmall training samples. However, it used a lot of trans-

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former structure; the large number of parameter counts of Vit increases the risk of over-fitting on remote sensing data sets with limited annotation samples [22]. Ma et al. [23] introduced the interactive spatial–spectral perception network (ISSP-Net). It addressed the challenge of optimizing the complementary information between PAN and MS images by enhancing spatial location interaction and integrating multiscale context through pixel-guided spatial enhancement. Additionally, Lu et al. [24] argued that current fusion techniques often depend on cascaded or weighted approaches, and that pixel-level feature fusion typically underperforms in classification. Therefore, they introduced a coupled adversarial learning-based classification (CALC). It involved training a subnetwork to extract high-level semantic features from multisource remote sensing data without supervision. Nevertheless, these methods do not take into account the cross-modal feature alignment, which makes it difficult to explore the common semantic structure of the model, and it is difficult to build a unified joint representation space, which hinders feature fusion and classification.

Based on the problems of feature space misalignment and neglect of complex relationships between modalities in the above studies, the following issues remain to be addressed.

1) *Cross-Modal Feature Space Misalignment:* Current approaches largely overlook the cross-modal feature space alignment problem [25]. PAN and MS images, derived from disparate imaging physics, lead to a significant modality gap for semantically equivalent samples, meaning they reside in nonoverlapping regions of the high-dimensional feature space. This representation space mismatch forces the classifier to learn from feature streams with inconsistent distributions, violating the fundamental assumptions of its learning paradigm, thereby compromising the model's generalization capability and ultimately leading to performance degradation.
2) *Discordance Between Detail and Semantic Information:* Most methods process features at a single scale [26], [27], disrupting the hierarchical flow of information from fine-grained details to abstract semantics. Low-level features contain rich spatial structural information but lack semantic clarity, whereas high-level features possess clear semantic concepts but suffer from severe spatial detail erosion. This fragmentation of the multiscale information flow prevents the effective complementarity and synergy between detailed and semantic information, leading to a less discriminative final fused representation.
3) *Imbalance Between Modality-Specific (Peculiar) and Shared Features:* Over-emphasizing modality-peculiar characteristics impedes the discovery of shared semantic representations across modalities [28], [29], [30]; whereas excessive pursuit of cross-modal feature sharing (enforcing representation uniformity) may dilute the discriminative peculiarities essential for fine-grained classification, leading to negative transfer. As evidenced in Fig. 1, this peculiar-shared disparity manifests as persistent feature space separation at category boundaries.

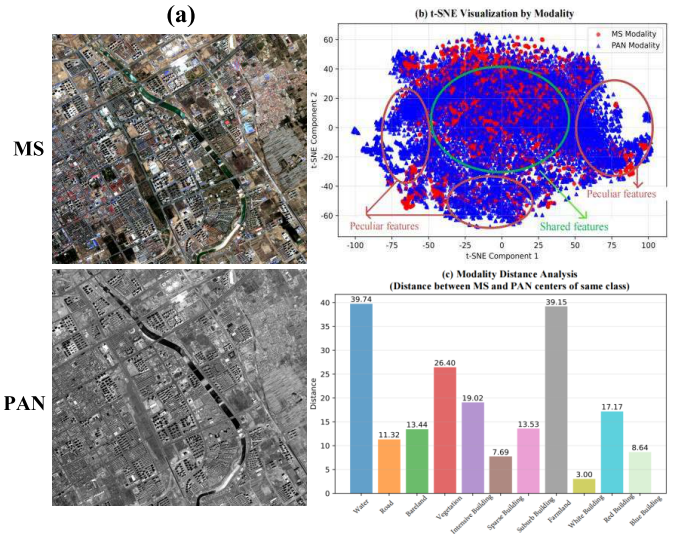


Fig. 1. Distribution characteristics of MS and PAN images in the original feature space on Hohhot dataset. (a) MS and PAN images. (b) Partial overlap in central regions but persistent separation at edges, reflecting intrinsic representational differences. (c) Quantifies the misalignment by measuring intermodal centroid distances per category, revealing significant cross-category variations.

To address the challenges posed by the integration of different modalities in remote sensing, this article proposes a hierarchical cross-modality aggregation network (HCMA-Net) to align, aggregate, and enhance features from multimodal images, thereby improving multimodal classification accuracy. The main contributions of this article are as follows.

1) To tackle the disparate feature representations of semantically equivalent samples across modalities, we propose HCMA-Net. It integrates cross-modal interactions through a four-stage fusion architecture, which is theoretically grounded in the concept of progressive distribution alignment. The HCMA-Net simultaneously and progressively adjusts heterogeneous spatial–spectral features from pixel-level to semantic-level through a nonlocal attention mechanism to minimize the feature distribution bias between modalities. This process forces intermodal sharing of high-dimensional semantics in a unified latent space, effectively bridging the representation gap.
2) To overcome the limitations imposed by a single scale and to take advantage of cross-modal interactions across scales, we design a hierarchical feature aggregation and enhancement module (HFAE-Module). Its core innovation is the hierarchical cross-modality feature aggregation (HCMFA) mechanism, which is designed to model cross-scale contextual dependencies. It utilizes cross-spectral nonlocal attention (CSNLA) to achieve spectral consistency by correlating long-range band-wise interactions, and spatial aggregation nonlocal attention (SANLA) to achieve spatial coherence by building dense correspondences across modalities. This dual-attention design enables feature interaction across different scales, effectively reducing the inherent differences in heterogeneous data and resolving the discordance between

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high-resolution detail and low-resolution semantic information.

3) In order to address the disparity between modality-peculiar and shared feature learning, this article presents a cross-modality interactive feature extraction module (CMIFE-Module). Its design is grounded in the information bottleneck principle, which seeks to learn a compressed yet informative representation. It employs self-attention to reinforce peculiar feature coherence and preserve task-relevant discriminative characteristics within each modality. Simultaneously, it utilizes cross-attention to facilitate a selective exchange of information between modalities that prioritizes the most complementary shared representations while filtering out redundant or noisy peculiarities. This mitigates feature loss occasioned by the peculiar-shared imbalance and effectively navigates the tradeoff between preserving modality-specific peculiarities and achieving cross-modal shared representations, thereby avoiding negative transfer.

The subsequent sections of this article are structured as follows. Section II reviews the work related to our designed approach. Section III outlines our designs in depth. Section IV introduces the dataset used in our research and analyzes the experimental results. Finally, Section V contains the article's conclusion.

II. Related Work

A. Hierarchical Feature Learning

The characteristic of having multiple layers inherent in remote sensing images has led to the development of frameworks that can extract features across different scales. Several studies have adopted pyramid-based architectures [31] to aggregate features at different spatial resolutions. Yan et al. [32] extended this idea with skip connections to preserve fine-grained details. Wang et al. [33] proposed a multiscale interactive fusion network (MIFNet), which improved the accuracy and efficiency of multisource remote sensing data classification through the extraction of multiscale interactive information and the fusion of global dependencies. More recently, transformer-based approaches [34], [35] have demonstrated superior performance in modeling long-range dependencies across scales. The SpectralFormer [36], which was based on the Transformer model, could learn local spectral information from neighboring bands and generate grouped spectral embeddings. Liu et al. [37] introduced OSHFNet that employed CNNs for optical image local feature extraction and VMamba enhanced with dynamic snake convolution (DSConv) for SAR image global structural learning, achieving hierarchical feature integration through a multiscale dynamic gating fusion module that adaptively aligned spectral-spatial representations across modalities. Nonetheless, current hierarchical methodologies process diverse modalities in isolation prior to fusion, thereby overlooking the potential for cross-scale interactions between modalities during the feature extraction phase.

B. Cross-Modality Alignment

Recent work has shown that explicitly modeling these distributional mismatches through attention mechanisms can significantly improve fusion quality [38], [39]. Yang et al. [40] proposed a correlation alignment loss to minimize distribution discrepancies between modalities. Alternatively, Hu et al. [41] employed adversarial learning to achieve domain-invariant feature representations. The CMX [42] introduced a cross-modal feature rectification module that significantly improved generalization across diverse sensor types. Ma et al. [43] introduced a cross-modal spatial–spectral interaction mamba (CMS^2I-Mamba) that achieved efficient spatial–spectral fusion through adaptive scanning and cross-modal alignment. While these approaches demonstrate promising results, they often lack explicit mechanisms for preserving modality-specific characteristics during alignment, potentially leading to information loss.

III. Methodology

In this chapter, the HCMA-Net for multimodal classification is introduced in Section III-A, which includes the HFAE-Module in Section III-B, and the CMIFE-Module in Section III-C.

A. Hierarchical Cross-Modality Aggregation Network

For multimodal data, processing the data of the respective modalities through a multibranch network structure can suffer from feature space misalignment. Therefore, we present the HCMA-Net in Fig. 2, which consists of four main stages for extracting, aggregating, and refining the features step by step.

The inputs are multispectral (MS, $I_{\mathrm{MS}} \in \mathbb{R}^{B\times 4\times 16\times 16}$) and panchromatic (PAN, $I_{\mathrm{PAN}} \in \mathbb{R}^{B\times 1\times 64\times 64}$) images, respectively. B presents the batch size. In order to align the two modalities in feature dimension for subsequent processing, we first perform independent encoding operations on them.

1) *MS Image Encoding:* Considering the relatively low spatial resolution of MS image, we use Transposed Convolution to up-sample and transform the feature dimensions

$$F_{\mathrm{MS}_{\mathrm{encoded}}} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{ConvTranspose2d}\left(I_{\mathrm{MS}}\right)\right)\right) \quad (1)$$

where ConvTranspose2d(·) adjusts the number of channels of input I_{MS} from 4 to 64, and also upscale its spatial resolution to align with the encoded PAN image. The BN(·) stands for BatchNorm and ReLU(·) is the activation function. Output features $F_{\mathrm{MS}_{\mathrm{encoded}}} \in \mathbb{R}^{B\times 64\times 64\times 64}$.

2) *PAN Image Encoding:* PAN image have high spatial resolution, and we use standard convolutional layers for feature extraction and dimensionality adjustment

$$F_{\mathrm{Pan}_{\mathrm{encoded}}} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv2d}\left(I_{\mathrm{PAN}}\right)\right)\right) \quad (2)$$

where Conv2d(·) adjusts the number of channels of the input I_{PAN} from 1 to 64, and matches the $F_{\mathrm{MS}_{\mathrm{encoded}}}$ with spatial dimensions. Output features $F_{\mathrm{Pan}_{\mathrm{encoded}}} \in \mathbb{R}^{B\times 64\times 64\times 64}$.

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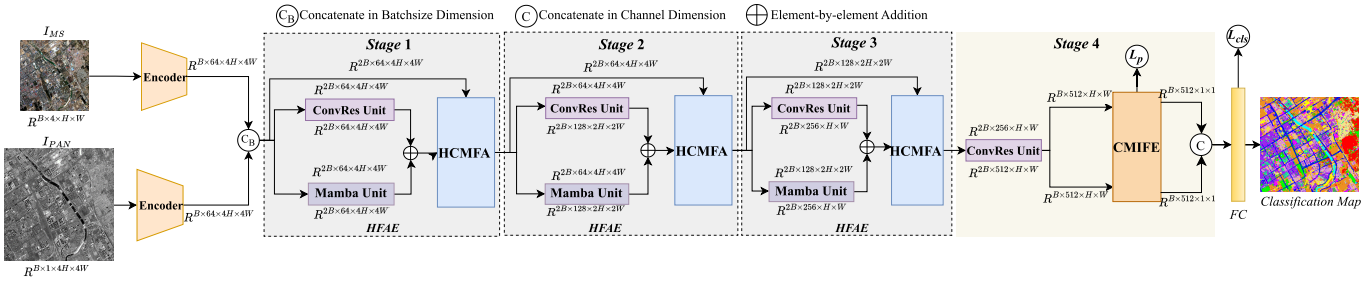


Fig. 2. Pipeline of the HCMA-Net. HCMFA represents the hierarchical cross-modality feature aggregation. CMIFE represents the cross-modality interactive feature extraction. H, W, B represent height, width, and batch size, respectively. (Top) Each box is the input dimension and (bottom) output dimension.

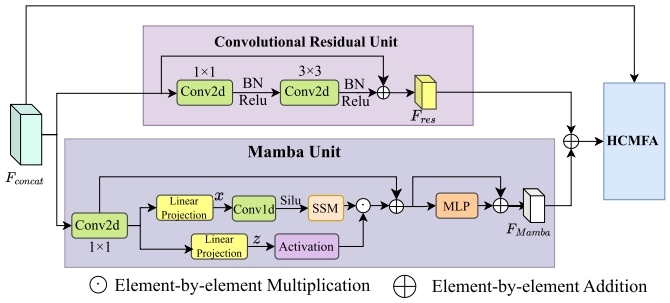


Fig. 3. Illustration of the HFAE. The specific process of HCMFA is shown in Fig. 4.

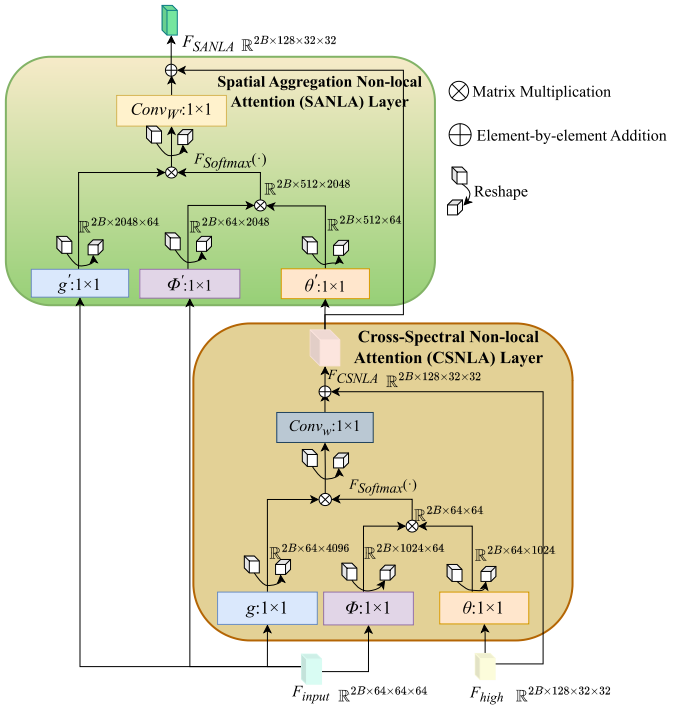


Fig. 4. Illustration of HCMFA, which includes the CSNLA layer and SANLA layer. To fully demonstrate the process of cross-scale cross-modal feature aggregation, we take stage 2 as an example and annotate the variable dimensionality changes at key steps.

After initial encoding and alignment, we concatenate the feature maps ($C_B(\cdot)$) of the two modalities along the batch dimension

$$F_{\text{concat}} = C_B\left(\left[F_{\text{MS}_{\text{encoded}}}, F_{\text{Pan}_{\text{encoded}}}\right]\right). \tag{3}$$

Concatenated feature map $F_{\text{concat}} \in \mathbb{R}^{2B\times 64\times 64\times 64}$. This splicing allows subsequent feature extraction layers to process information from both modalities simultaneously, facilitating the learning of shared feature representations.

Subsequently, F_{concat} is fed into a multistage feature extraction backbone network.

1) *Stages 1–N (According to the Experiments in Section IV-B, N is Taken to be 3):* The HFAE-Module in Fig. 3 (In Fig. 3, HCMFA is shown in Fig. 4) is responsible for extracting and aggregating features from multimodal images, facilitating effective modality alignment.
2) *Stage 4:* The CMIFE-Module in Fig. 5 enhances the complementarity and coherence of the integrated features through sophisticated interaction and attention mechanisms, thereby augmenting the final classification performance.

The loss function of the HCMA-Net consists of two parts, one for the cross-entropy loss [44] ($\mathcal{L}_{\text{cls}}$) required for the classification task and the other for the complementary representation supervised learning loss [45] to optimize the effect of feature alignment ($\mathcal{L}_p$).

The $\mathcal{L}_p$ adaptively penalizes feature redundancy by dynamically weighting coefficients derived from information entropy and feature similarity. Utilizing a Sigmoid-based penalty mechanism, it selectively enhances complementary learning for dissimilar features while balancing processing for similar ones, effectively mitigating information redundancy in multisource classification

$$\mathcal{L}\text{oss} = \alpha\mathcal{L}_p + \mathcal{L}_{\text{cls}} \tag{4}$$

where α[1] is the weight coefficient.

Through the gradual augmentation of feature representation, HCMA-Net is adeptly equipped to tackle the complexities inherent in multimodal feature fusion.

B. Hierarchical Feature Aggregation and Enhancement Module

Different imaging techniques for PAN and MS images result in significant visual and physical property differences.

[1]In this article, according to Section IV-B, α is 0.5.

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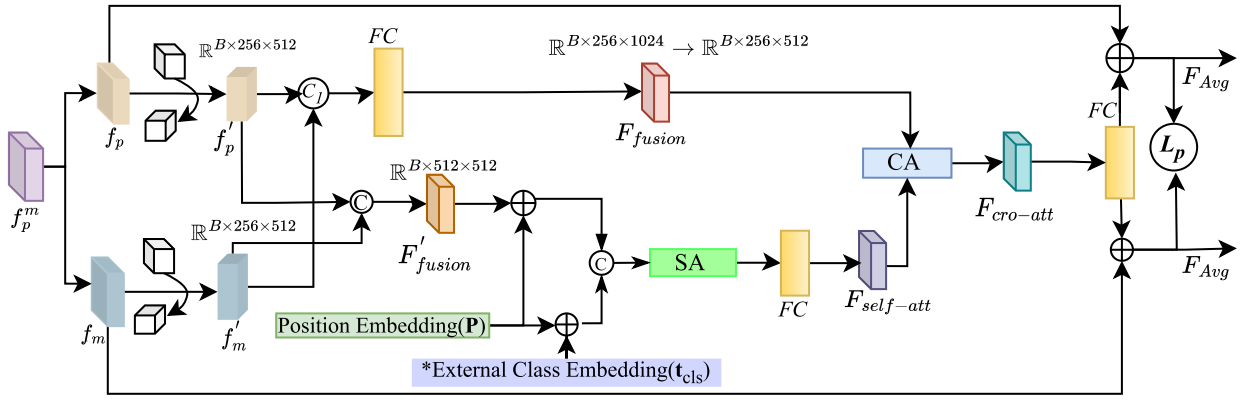


Fig. 5. CMIFE-Module. SA and CA represent the self-attention and cross-attention based on the transformer. ⓒ represents the concatenation in the last dimension. The two F_{Avg} denote GAP operations.

Single-scale alignment of spatial–spectral features cannot bridge the gap between the detailed information of high resolution and the semantic information of low resolution, disrupting the continuity of detail-semantic information. Therefore, we propose the HFAE-module in Fig. 3 to align the feature space between different modalities progressively. Each HFAE-Module consists of three components: a convolutional residual unit, a Mamba unit, and the HCMFA. Note that the HFAE-Module contains a total of three stages, and we will only take the second stage as an example.

1) Parallel Feature Extraction: The HFAE-Module contains two parallel branches, the convolutional residual unit and the Mamba unit, based on the state-space model. These two paths process the features at both local and global levels, respectively.

a) Convolutional Residual Unit: Convolutional Residual Unit [46] extracts features by using convolution, batch normalization, and ReLU activation functions. Residual connection helps preserve the original information and mitigates the problem of vanishing gradients.

The feature aggregation process is performed hierarchically across three stages. It is critical to note that the input feature F_{concat} in (5) is stage-dependent: In Stage 1, it is the initial fused feature from (3); In Stages 2 and 3, it represents the output from the previous stage's HCMFA module. This design enables the progressive refinement and alignment of features

$$F_{\text{res}} = \text{ReLU}\left(\text{BN}\left(W_2 * \left(\text{ReLU}\left(\text{BN}\left(W_1 * F_{\text{concat}}\right)\right)\right)\right)\right) + F_{\text{concat}} \tag{5}$$

where W_1 and W_2 are learnable parameters, and $F_{\text{res}} \in \mathbb{R}^{2B\times128\times32\times32}$ is the output of the convolutional residual unit.

b) Mamba Unit: To overcome the limitation of convolutional operations in modeling the global context, we incorporate the Mamba block. As an efficient alternative to Transformers, Mamba utilizes a selective state space model to achieve powerful long-range dependency modeling with only linear computational complexity $O(N)$ [47], [48]. Its input-dependent mechanism allows for dynamic feature selection, making it highly effective and efficient for handling the long sequences characteristic of high-resolution multimodal remote sensing imagery

$$\begin{aligned} F_{\text{Mamba}} &= \text{Mamba}\left(\text{Conv2d}\left(F_{\text{concat}}\right)\right) \\ F_{\text{high}} &= F_{\text{Mamba}} + F_{\text{res}} \end{aligned} \tag{6}$$

where Mamba$(\cdot)$ denotes the Mamba [49] architecture. F_{high} and F_{Mamba} have the same dimension as F_{res}.

2) Hierarchical Cross-Modality Feature Aggregation (HCMFA): As shown in Fig. 4, the HCMFA is responsible for aligning and aggregating features in the channel and spatial dimensions of the features to ensure effective complementarity between different features. We take stage 2 in Fig. 2 as an example to show the specific process of cross-scale, cross-modal feature alignment.

The HCMFA mechanism at each stage takes two inputs: 1) the current stage's deep, downsampled features ($F_{\text{Mamba}} + F_{\text{res}}$, resolution 32×32); and 2) the high-resolution output from the previous stage (F_{concat}, resolution 64×64). This allows for cross-scale feature interaction within the module.

a) CSNLA Layer: The goal of the CSNLA Layer is to align spectral features of different modalities through interactions in feature space. It can first establish spectral consistency early in the feature to reduce the spectral differences between different modalities. Initially, the shallow ($F_{\text{input}} \in \mathbb{R}^{2B\times64\times64\times64}$) and deep features ($F_{\text{high}} \in \mathbb{R}^{2B\times128\times32\times32}$) are subjected to a series of 1×1 convolutional transformations, denoted as $g(\cdot), \theta(\cdot)$ and $\phi(\cdot)$ to generate the features g_x, ϕ_x, and θ_x

$$\begin{aligned} g_x &= g\left(F_{\text{input}}\right) \\ \phi_x &= \phi\left(F_{\text{input}}\right) \\ \theta_x &= \theta\left(F_{\text{high}}\right). \end{aligned} \tag{7}$$

To match the dimensionality requirements of matrix multiplication, the dimensions of g_x, ϕ_x and θ_x are adjusted to be $\mathbb{R}^{2B\times64\times4096}$, $\mathbb{R}^{2B\times1024\times64}$, $\mathbb{R}^{2B\times64\times1024}$, respectively.

Subsequently, we perform matrix multiplication of θ_x and ϕ_x and normalize the result to obtain the attention weight $A_{\text{spe}} \in \mathbb{R}^{2B\times64\times64}$.

In conclusion, the weighted high-dimensional features obtained by matrix multiplication of A_{spe} and g_x are

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reconverted ($\mathrm{Conv}_w(\cdot)$, 1×1 convolution) to match the dimensionality of the original inputs. The transformed features are aggregated with the original high-dimensional features F_{high} to achieve spectral aggregation ($F_{\text{CSNLA}} \in \mathbb{R}^{2B\times128\times32\times32}$)

$$\begin{aligned} A_{\text{spe}} &= \theta_x \times \phi_x \\ F_{\text{CSNLA}} &= \mathrm{Conv}_w\left(A_{\text{spe}} \times g_x\right) + F_{\text{high}}. \end{aligned} \tag{8}$$

b) SANLA Layer: The SANLA layer is employed to ensure the preservation of spatial structures, utilizing a nonlocal mechanism to align spatial details from both modalities. This architecture bears a resemblance to the CSNLA layer.

The CSNLA layer focuses on capturing long-range spectral correlations between PAN and MS images, while the SANLA layer focuses on capturing spatial relationships and aggregating spatial features, which helps to focus on pixel-level information rather than coarser region- or channel-level information, thus addressing the spatial-spectral misalignment. By aggregating features in both spectral and spatial dimensions with different scales, HCMFA reduces the risk of feature space misalignment and improves classification accuracy.

C. Cross-Modality Interactive Feature Extraction Module

The balance between peculiar and shared features is pivotal. While overemphasizing peculiar specificity leads to noisy overfitting, excessive intermodal alignment erases discriminative details. Therefore, we propose the CMIFE-Module, which is used in the fourth stage of the HCMA-Net in Fig. 5 to enhance cross-modal relationships while maintaining the integrity of the peculiar features. Before the CMIFE-Module, the output of stage 3 will be further passed through the convolutional residual unit to obtain a deeper representation of the features $f_p^m \in \mathbb{R}^{2B\times512\times16\times16}$.

$f_p^m \in \mathbb{R}^{2B\times512\times16\times16}$ are separated to obtain f_p and $f_m \in \mathbb{R}^{B\times512\times16\times16}$. f_p and f_m are reshaped into a sequence that is appropriate for the subsequent attention operations as $f_p', f_m' \in \mathbb{R}^{B\times256\times512}$.

In order to promote efficient interaction among distinct features, f_p' and f_m' are integrated along the last dimension ($\mathcal{C}_1(\cdot)$) and subjected to a fully connected layer ($\mathrm{FC}(\cdot)$) to derive $F_{\text{fusion}} \in \mathbb{R}^{B\times256\times512}$

$$F_{\text{fusion}} = \mathrm{FC}\left(\mathcal{C}_1\left(f_p', f_m'\right)\right). \tag{9}$$

To construct the peculiar interaction features, we first splice f_p' and f_m' along the sequence length dimension ($f_p', f_m' \in \mathbb{R}^{B\times256\times512} \rightarrow F'_{\text{fusion}} \in \mathbb{R}^{B\times512\times512}$). Positional information is crucial for maintaining spatial relationships within the fused features. Therefore, learnable positional embedding parameters ($\mathbf{P} \in \mathbb{R}^{1\times513\times512}$) are added to ensure spatial consistency.

Then, in addition to the interaction features, a learnable class token ($\mathbf{t}_{\text{cls}} \in \mathbb{R}^{1\times1\times512}$) is introduced, which serves as a global representative of the combined features. Specifically, F'_{fusion} is transformed by applying positional encoding ($\mathbf{P}_{[:,1:]}$), which embeds spatial relationships into the feature representation. Simultaneously, $\mathbf{t}_{\text{cls}}$ is augmented with positional information from the first position embedding ($\mathbf{P}_{[:,:1]}$). It serves as a global representation that aggregates feature contextual cues. The class token is concatenated ($C(\cdot,\cdot)$) to the feature sequence

$$\begin{aligned} F_{\text{interaction}} &= F'_{\text{fusion}} + \mathbf{P}_{[:,1:]} \\ \mathbf{T}_{\text{cls}} &= \mathbf{t}_{\text{cls}} + \mathbf{P}_{[:,:1]} \\ F^{\text{cls}}_{\text{interaction}} &= C\left(\mathbf{T}_{\text{cls}}, F_{\text{interaction}}\right) \end{aligned} \tag{10}$$

where $F_{\text{interaction}} \in \mathbb{R}^{B\times512\times512}$, $\mathbf{P}_{[:,1:]} \in \mathbb{R}^{1\times512\times512}$, $\mathbf{P}_{[:,:1]} \in \mathbb{R}^{1\times1\times512}$, $\mathbf{T}_{\text{cls}} \in \mathbb{R}^{B\times1\times512}$, $F^{\text{cls}}_{\text{interaction}} \in \mathbb{R}^{B\times513\times512}$.

Additionally, the feature interaction process is facilitated through a four-head self-attention mechanism and $\mathrm{FC}(\cdot)$ applied to $F^{\text{cls}}_{\text{interaction}}$, which allows each feature to consider the relationships with all other spatial positions within the different modalities

$$F_{\text{self-att}} = \mathrm{FC}\left(\mathrm{softmax}\left(\frac{QK^T}{\sqrt{128}}\right)V\right) \tag{11}$$

where $Q = F^{\text{cls}}_{\text{interaction}}W_Q$, $K = F^{\text{cls}}_{\text{interaction}}W_K$, $V = F^{\text{cls}}_{\text{interaction}}W_V$, W_Q, W_K and W_V are feature embeddings, $F_{\text{self-att}} \in \mathbb{R}^{B\times513\times512}$.

Subsequently, a four-head cross-attention is applied to refine the integration between features further to get $F_{\text{cro-att}} \in \mathbb{R}^{B\times256\times512}$. In this stage, the fused features (F_{fusion}) act as the query (Q_{fusion}), while the self-attended features ($F_{\text{self-att}}$) provide the key (K_{inter}) and value (V_{inter})

$$F_{\text{cro-att}} = \mathrm{softmax}\left(\frac{Q_{\text{fusion}}K^T_{\text{inter}}}{\sqrt{128}}\right)V_{\text{inter}}. \tag{12}$$

The process reduces $F_{\text{cro-att}}$ to its initial spatial resolution ($F_{\text{CMIFE}} \in \mathbb{R}^{B\times512\times16\times16}$) while maintaining a constant number of channels via the fully connected layer, subsequently summing it with the original input to obtain $F^{\text{PAN}}_{\text{CMIFE}}$ and $F^{\text{MS}}_{\text{CMIFE}} \in \mathbb{R}^{B\times512\times16\times16}$

$$\begin{aligned} F_{\text{CMIFE}} &= \mathrm{FC}\left(F_{\text{cro-att}}\right) \\ F^{\text{PAN}}_{\text{CMIFE}} &= F_{\text{CMIFE}} + f_p \\ F^{\text{MS}}_{\text{CMIFE}} &= F_{\text{CMIFE}} + f_m. \end{aligned} \tag{13}$$

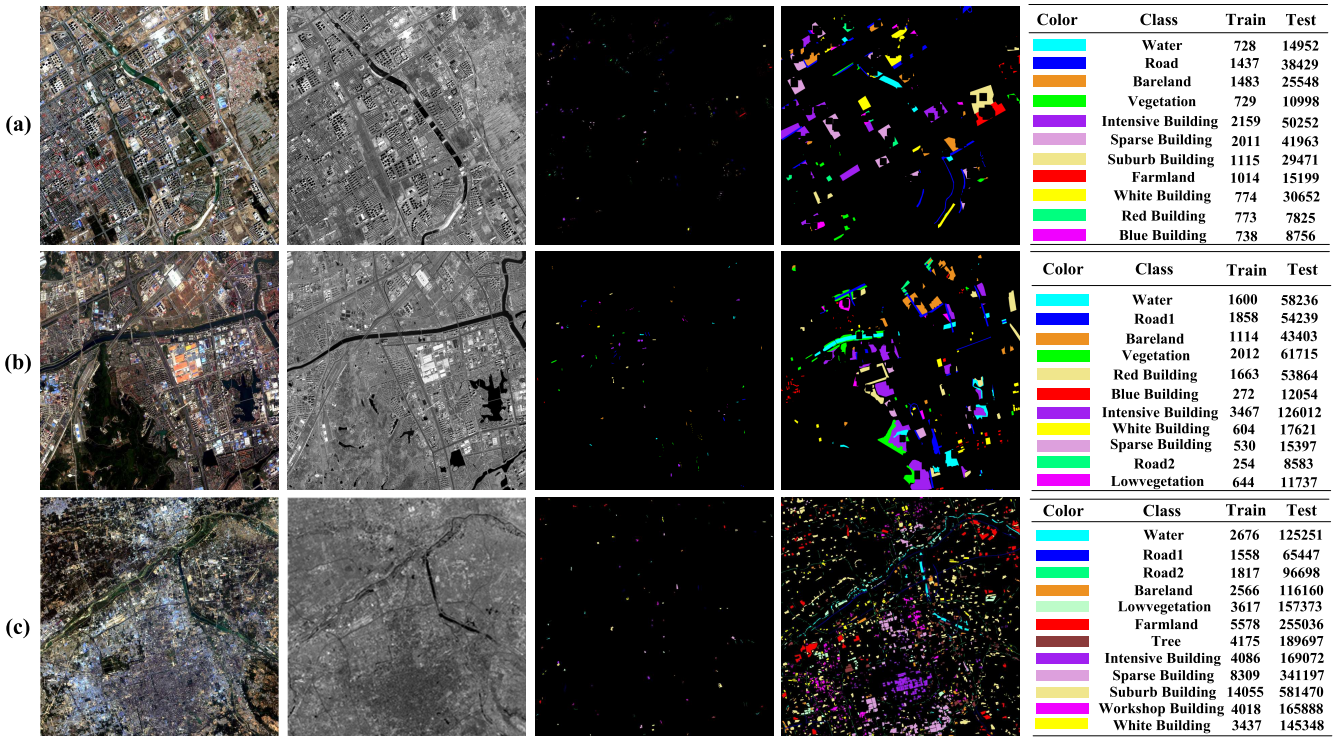
In conclusion, both $F^{\text{PAN}}_{\text{CMIFE}}$ and $F^{\text{MS}}_{\text{CMIFE}}$ will be entered into the complementary representation of the supervised learning loss ($\mathcal{L}_p$) [45] to promote complementary feature learning, thus generating a nonredundant multisource representation. Conversely, $F^{\text{PAN}}_{\text{CMIFE}}$ and $F^{\text{MS}}_{\text{CMIFE}}$ will be prepared for the final classification task by downscaling the feature map size through global average pooling (GAP).

The CMIFE-Module applies self-attention to enhance modality-specific features and cross-attention to promote effective information exchange between PAN and MS features, capturing their complementary characteristics more effectively. It yields a more discriminative feature representation, directly contributing to improved classification accuracy.

IV. Experimental Study

In this section, we provided a detailed overview of the dataset and experimental setup, hyperparameter selection, ablation experiments, and compared our proposed HCMA-Net with other methods in terms of classification performance and feature space alignment to evaluate its effectiveness and robustness.

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Color	Class	Train	Test
	Water	728	14952
	Road	1437	38429
	Bareland	1483	25548
	Vegetation	729	10998
	Intensive Building	2159	50252
	Sparse Building	2011	41963
	Suburb Building	1115	29471
	Farmland	1014	15199
	White Building	774	30652
	Red Building	773	7825
	Blue Building	738	8756

Color	Class	Train	Test
	Water	1600	58236
	Road1	1858	54239
	Bareland	1114	43403
	Vegetation	2012	61715
	Red Building	1663	53864
	Blue Building	272	12054
	Intensive Building	3467	126012
	White Building	604	17621
	Sparse Building	530	15397
	Road2	254	8583
	Lowvegetation	644	11737

Color	Class	Train	Test
	Water	2676	125251
	Road1	1558	65447
	Road2	1817	96698
	Bareland	2566	116160
	Lowvegetation	3617	157373
	Farmland	5578	255036
	Tree	4175	189697
	Intensive Building	4086	169072
	Sparse Building	8309	341197
	Suburb Building	14055	581470
	Workshop Building	4018	165888
	White Building	3437	145348

Fig. 6. Dataset is organized into three rows, listed from top to bottom as (a) Hohhot area, (b) Nanjing area, and (c) Xi'an area. The columns, arranged from left to right, comprise MS images, PAN images, training ground-truth images, testing ground-truth images, and their corresponding labels. Additionally, the PAN images have been resized to align with the dimensions of the MS images to enhance visualization.

A. Dataset Description and Experimental Settings

1) Datasets: Fig. 6(a)–(c) illustrates the panchromatic (PAN) and MS images of Hohhot, Nanjing, and Xi'an, respectively.

1) *Hohhot*: The dataset in Fig. 6(a) includes Level 1A imagery from the Gaofen-2 satellite over Hohhot, China, on May 23, 2015. The MS image is $2001 \times 2101 \times 4$ pixels with a 4 m resolution, and the PAN image is 8004×8404 pixels with a 1 m resolution. It covers 11 land cover categories: water, roads, bareland, vegetation, farmland, intensive building, sparse building, suburb building, white building, red building, and blue building.
2) *Nanjing*: Fig. 6(b) shows Level 1A imagery from the Gaofen-2 satellite over Nanjing, China, on April 21, 2015. The MS image has $2000 \times 2500 \times 4$ pixels with 8 m resolution; the PAN image has $8000 \times 10\,000$ pixels with 2 m resolution. The area is divided into 11 land cover categories: water, road1, bareland, vegetation, red and blue roof buildings, urban dense high-rise buildings, white roof buildings, urban sparse residential buildings, road2, and low vegetation.
3) *Xi'an*: As shown in Fig. 6(c), the MS image is a four-band image with a spatial resolution of 8 m and dimensions of $4548 \times 4541 \times 4$. The PAN image, with a spatial resolution of 2 m, is a single-band image measuring $18\,192 \times 18\,164 \times 1$ pixels. The ground truth image of the Xi'an scene comprises several land cover classes, including water bodies, arboreal vegetation, low-lying vegetation, bareland, farmland, road1, road2, intensive building, sparse building, suburb building, workshop building and white building.
4) *Houston2013*: As depicted in Fig. 7, the Houston 2013 dataset is composed of hyperspectral imagery (HSI) and a digital surface model (DSM) derived from LiDAR, with a spatial resolution of 349×1905 pixels. The HSI data encompasses 144 spectral bands, whereas the LiDAR data comprises a single spectral band.

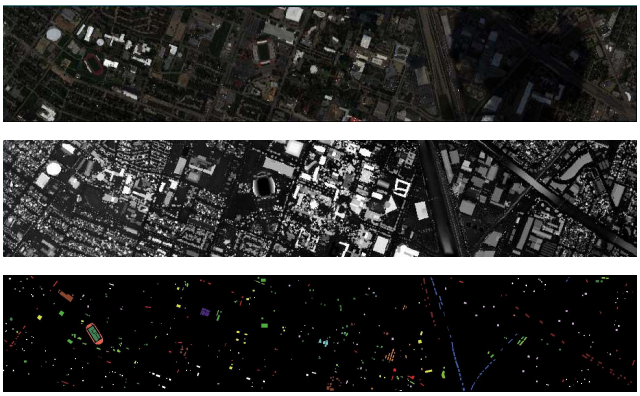


Fig. 7. Visual samples of the Houston 2013 dataset. (Top to bottom) Pseudo color visualizations of HSI data, a view of LiDAR data, and ground truth.

2) Experimental Environment: The experimental procedures of this study are executed on a computing platform

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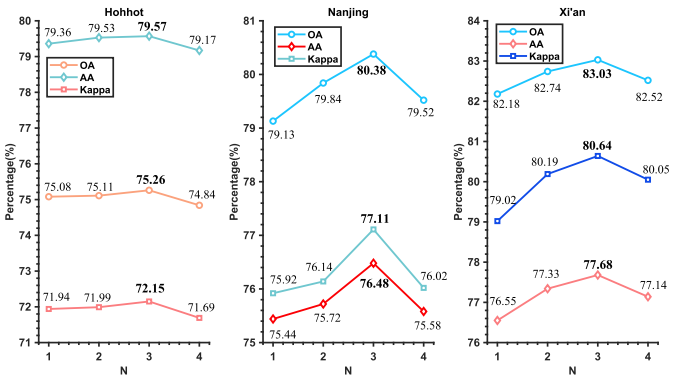


Fig. 8. Effects of the number of feature aggregation N on OA, AA, and Kappa in three datasets.

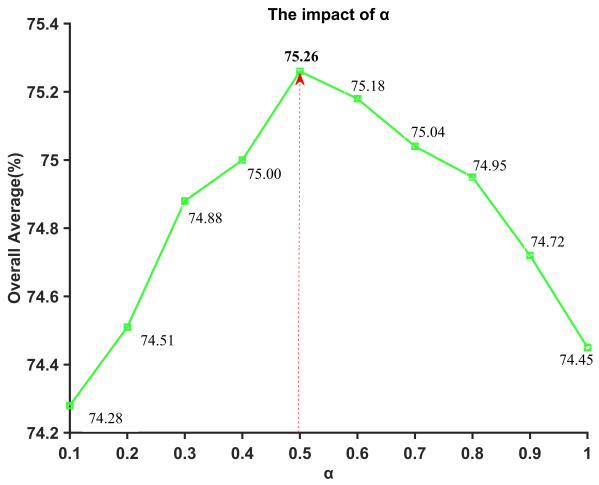


Fig. 9. Impact of α on OA in Hohhot dataset.

equipped with a 12-GB GPU of the RTX2080Ti and 128 GB of RAM. The network architecture is implemented and trained via the PyTorch framework. Throughout the training phase across all datasets, a weight decay parameter of 0.0005 is employed, the number of epochs is maintained at 30, and an initial learning rate of 0.001 is utilized. A batch size of 32 is consistently applied to all datasets, and the Adaptive Moment Estimation (Adam) algorithm is employed for optimization.

Given the 4:1 resolution ratio between the PAN and MS images, the dimensions for PAN input are set at $64 \times 64 \times 1$, whereas for MS, they are $16 \times 16 \times 4$. In the case of the HS and LiDAR images, the 1:1 resolution ratio shared by both leads to identical input sizes of 16×16 for each.

3) Evaluation Indicators: The performance and classification outcomes of HCMA-Net are evaluated through various metrics such as overall accuracy (OA), average accuracy (AA), and the kappa coefficient (Kappa). The computational complexity is measured in floating point operations (FLOPs), representing the total number of floating-point operations required for a single forward pass.

B. Selection of Hyperparameters

In this section, we present a comprehensive quantitative analysis of the hyperparameters using a controlled variable approach. All parameters are held constant, except for the specific hyperparameters under investigation.

1) *The Times of Feature Aggregation N:* Indeed, excessive aggregation of spatial and spectral features can produce a degree of feature interference, resulting in the masking or distortion of some subtle but important features. Insufficient aggregation will result in the fused features not being able to fully express the information of the original image, which reduces the expressive power of the features. Therefore, it is necessary to balance the effect of spatial and spectral feature aggregation precisely. As shown in Fig. 8, when N is low, the HCMA-Net cannot effectively fuse the spatial details with the spectral information, resulting in the final classification results that cannot simultaneously combine high spatial resolution and rich spectral features. Conversely, over-aggregation may lead to a large amount of duplicated information between spatial and spectral features, increasing computational complexity without providing more valid information. Therefore, the number of times of feature aggregation in all the experiments in this article is set to 3.
2) *α in (4):* The α parameter significantly influences the degree of modal alignment, thereby reducing substantial information redundancy across data sources while preserving the unique information pertinent to each individual source. This process facilitates the exclusion of information that is noncontributory to classification accuracy. As depicted in Fig. 9, optimal classification performance is achieved when α is set to 0.5. Deviations of α from the value of 0.5 result in pronounced fluctuations of the OA value. Consequently, we employ the optimal split ratio of $\alpha = 0.5$ for HCMA-Net in our subsequent experiments.
3) *Comparison and Exploration of Patch Size:* The size of the patch determines the amount of information in the neighborhood of the sample, which has an important impact on the classification of the center pixel. Therefore, it is particularly important to choose the appropriate patch size. It can be seen from Table I that when the patch size is large, the running time increases accordingly. If the patch is too small, the neighborhood information is insufficient, resulting in too few features being extracted. Conversely, if the patch size is too large, it causes redundancy in the neighborhood information. This redundancy introduces irrelevant interference, increasing the network calculation cost and affecting classification accuracy. Based on the analysis of the above experimental results, MS's patch size is finally determined to be 16, while PAN's patch size is 64.

C. Experiments With Different Cross-Modal Input Sources

Within the CMIFE-Module, the variation in input sources plays a critical role in dictating the efficacy of cross-modal interactions. Existing methodologies for cross-modal

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TABLE I
EXPERIMENTAL RESULTS OF HYPERPARAMETER PATCH SIZE

Dataset	Hohhot				Nanjing				Xi'an			
Patch Size	8	16	24	32	8	16	24	32	8	16	24	32
OA	74.66	**75.26**	74.33	74.42	79.86	**80.38**	80.01	79.66	82.06	**83.03**	82.11	82.33
AA	79.02	**79.57**	78.62	78.91	76.04	**76.48**	76.22	75.63	76.81	**77.68**	76.89	77.02
Kappa	71.69	**72.15**	71.24	71.48	76.52	**77.11**	76.84	76.41	79.30	**80.64**	79.52	79.68
Time	142.09	238.74	302.55	371.04	205.68	254.11	320.88	406.37	1159.82	1458.46	1758.67	2183.41

TABLE II
DIFFERENT CROSS-MODAL INPUT SOURCES ON THE CMIFE-MODULE IN HOHHOT DATASET

No.	Query(Q), Key(K), Value(V) Configuration	OA	AA	Kappa	Flops(G)
(1)	$Q \rightarrow F_{fusion}$, $K/V \rightarrow F_{self-att}$	**75.26**	**79.57**	**72.15**	35.04
(2)	$Q \rightarrow F_{self-att}$, $K/V \rightarrow F_{fusion}$	74.75	78.89	71.62	35.04
(3)	(1) + (2)	74.88	79.02	71.77	42.06

TABLE III
ABLATION STUDY ON THE HCMA-NET IN HOHHOT DATASET

	Baseline	Mamba Unit	HFAE	CMIFE	OA(%)	AA(%)	Kappa(%)	Time(s)
(a)	✓				70.43	73.21	66.66	95.81
(b)	✓	✓			73.69	79.06	70.48	128.12
(c)	✓		✓		74.20	79.21	71.00	142.06
(d)	✓			✓	74.79	77.99	71.53	167.55
(e)	✓	✓	✓		74.94	**79.73**	71.81	163.37
(f)	✓	✓		✓	74.11	77.01	70.75	188.52
(g)	✓		✓	✓	74.59	78.92	71.43	215.68
(h)	✓	✓	✓	✓	**75.26**	79.57	**72.15**	238.74

TABLE IV
ABLATION STUDY ON THE CSNLA AND SANLA IN HFAE ON HOHHOT DATASET

CSNLA	SANLA	OA	AA	Kappa
✓		74.55	78.72	71.26
	✓	74.38	78.62	71.02
✓	✓	**75.26**	**79.57**	**72.15**

interaction attention typically involve three input scenarios: the first scenario utilizes F_{fusion} as the Query (Q), while employing $F_{\text{self-att}}$ as the Key (K) and Value (V). In the second scenario utilizes the $F_{\text{self-att}}$ as the Q, while employing F_{fusion} as the Key (K) and Value (V). The third scenario amalgamates the two configurations to thoroughly address the complexities of cross-modal interaction circumstances. Consequently, we undertake experimental analysis to ascertain the scenario that most effectively aligns with our requirements. According to Table II, F_{fusion} as a Q can effectively guide cross-modal information interactions and can effectively utilize semantic information in the deep network. The Dual Cross-Attention doesn't achieve better classification results, which may be due to the generation of redundant information in the process of bidirectional information complementarity.

D. Ablation Study: The Establishment of Network Architecture

This section studies the contribution of designed modules [CMIFE, hierarchical feature aggregation and enhancement (HFAE)] compared to the baseline. We preserve identical hyperparameter values and consistent experimental configurations throughout the process. Our baseline model is I^3RL-Net [45]. According to Table III, we can get the following conclusions.

1) Compared to baseline (a), when we add one of the components of Mamba Unit (b), HFAE (c), and CMIFE (d) individually, all of them can effectively enhance the feature extraction capability (70.43% versus 73.69%, 70.43% versus 74.20%, 70.43% versus 74.79%).
2) *Mamba Unit + HFAE (e):* This combination gives better results with OA of 74.94%, AA reaching 79.73%, and Kappa 71.81%. This suggests that there is a synergistic effect between the two modules, especially in improving the AA.
3) *Mamba Unit + CMIFE (f):* Performance is slightly degraded (74.11%), but still better than using either module alone. This may indicate some functional overlap between the two modules.
4) *HFAE + CMIFE (g):* Outperforms either module alone, suggesting that feature aggregation and peculiar and shared information balancing are important for the classification task.
5) *Combination of all Modules (h):* The best performance of the model is achieved when combining all the modules (HFAE + CMIFE). OA improved to 75.26%, AA to 79.57%, and Kappa to 72.15%. This demonstrates that individual modules can be most effective when fully integrated, with each module contributing uniquely to the final performance.
6) HCMFA consists of two parts, CSNLA and SANLA. As shown in Table IV, both CSNLA and SANLA layers have a significant impact on the classification results. This indicates that feature aggregation can be performed from both spectral and spatial perspectives.
7) As shown in Fig. 10, both CSNLA and SANLA can effectively enhance the local spatial structure information and alleviate the misalignment of feature space. CMIFE can effectively enhance the features after preliminary aggregation, so that singular features and common features can be effectively balanced, and then enhance the expression ability of features.
8) As shown in Fig. 11, compared with the baseline method, our method achieves a reduction of intraclass spacing while increasing interclass differences by aligning the eigenspaces between two modalities, which in turn effectively improves the performance.
9) Mamba Unit has a relatively low increase in reasoning time due to its linear complexity. HFAE is used in the first three stages of the network. With the deepening

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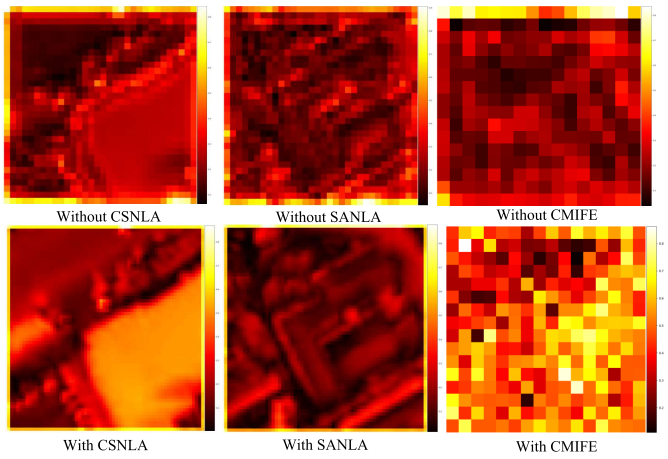


Fig. 10. Visualization results of intermediate feature maps.

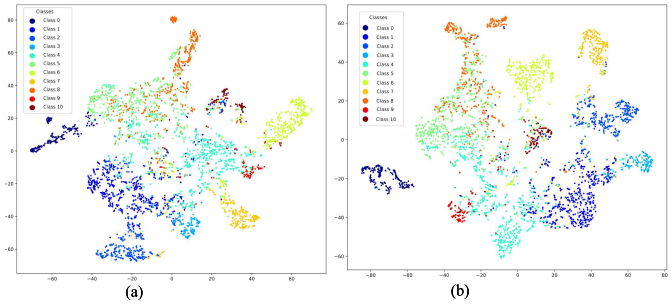


Fig. 11. Feature visualizations on Hohhot dataset. The horizontal and vertical axes represent the two main components of t-SNE algorithm after reducing high-dimensional data to 2-D space. (a) Baseline. (b) HCMA-Net.

of the network structure, its calculation dimension will also decrease, which makes its reasoning time shorter than CMIFE, but longer than the Mamba unit. Although CMIFE is only used in the fourth stage, it significantly uses multihead self-attention and cross attention mechanisms to enhance feature representation, resulting in a significant increase in computational complexity, which leads to a significant increase in reasoning time.

E. Experiment Results and Comparison Algorithms

In three datasets, the proposed HCMA-Net is compared with six state-of-the-art methods, namely MFT [50], GCFNet [20], CRHFF [51], AM^{3}Net [52], CALC [24], ISSP-Net [23], and CMS^{2}I-Mamba [43].

1) Comparison Results on the Hohhot Image: As shown in Table V and Fig. 12, the MFT enhances the transformer encoder module, enabling it to capture long-range feature dependencies more effectively. This architecture is particularly well-suited for processing multichannel features in HSI. However, its capacity to extract fine-grained texture information from full-color images is limited. Consequently, the model's performance is suboptimal in categories characterized by distinctive textural features, such as c_4 (Vegetation, 69.03%) and c_7 (Suburb Building, 67.83%).

Compared to the MFT, for the GCFNet, the recognition ability is obviously insufficient when facing the situation where the neighboring pixels are of different categories, which makes the recognition results in the c_{10} (Red Building, 32.37%) and c_{11} (Blue Building, 32.03%) categories inferior to those of other methods, which in turn leads to a significant decrease in the OA, AA, and Kappa values.

The CRHFF framework leverages integrated features ranging from shallow to deep and global to local, extracted from high-resolution MS and PAN imagery. It facilitates the extraction and fusion of cross-resolution latent information, enabling more robust modeling of imaged objects through multimodal representation, which enables significant improvement in classification accuracy on c_6 (Blue Building, 91.60% versus 44.95% versus 95.59%). It also has the lowest number of FLOPs (2.44 GFLOPs) with an input size of 16×16 and 64×64.

The AM^{3}Net demonstrates superior classification performance in categories c_7 (Suburb Building, 86.66%). It utilizes an adaptive and multiscale guided learning process that operates on hierarchical spectral and spatial data, aiming to produce more fitting and distinctive features tailored to the particular targets.

In contrast to the aforementioned models, the CALC incorporates a supervised multilevel feature fusion classification (MFFC) subnetwork, which is designed to learn features at low, mid, and high levels. It generates a spectrum of probability estimates for various classes based on the acquired features. These estimates are subsequently amalgamated to produce a refined classification outcome. As a result, the CALC achieves superior classification accuracy in several categories, notably c_1 (Water, 96.91%), c_3 (Bareland, 95.02%), and c_5 (Intensive Building, 65.52%).

The ISSP-Net compensates for the shortcomings in other methods for single-image feature extraction by combining the advantages of different modal images and each other's complementary features. As a result, it achieves the best classification result in category c_{11} (Blue Building, 78.13%).

The CMS^{2}I-Mamba efficiently captures long-range global dependencies with linear computational complexity. Within this architecture, the MPS^{2}M module employs adaptive spatial and spectral scanning strategies to selectively enhance task-relevant classification cues. Meanwhile, the CIAM leverages diversified channel interaction mechanisms to facilitate cross-channel information propagation and integration, thereby improving alignment across heterogeneous modalities. Furthermore, the semantic-aware calibration module (SACM), drawing inspiration from cross-attention principles, refines singular modality features through fused representations, mitigating the risk of complementary information loss during intermodal transmission. Therefore, it achieves the best classification result in category c_4 (Vegetation, 88.28%) and c_6 (Sparse Building, 74.02%).

Taking the reduction of modal differences as the starting point, the HCMA-Net constructs HFAE-Module and CMIFE-Module to effectively reduce modal differences caused by different imaging techniques. It ensures that the information of two modalities can be fully utilized, avoiding information loss. Therefore, the HCMA-Net achieves the best classification results on classes c_2 (Road, 86.79%), c_8 (Farmland, 95.86%),

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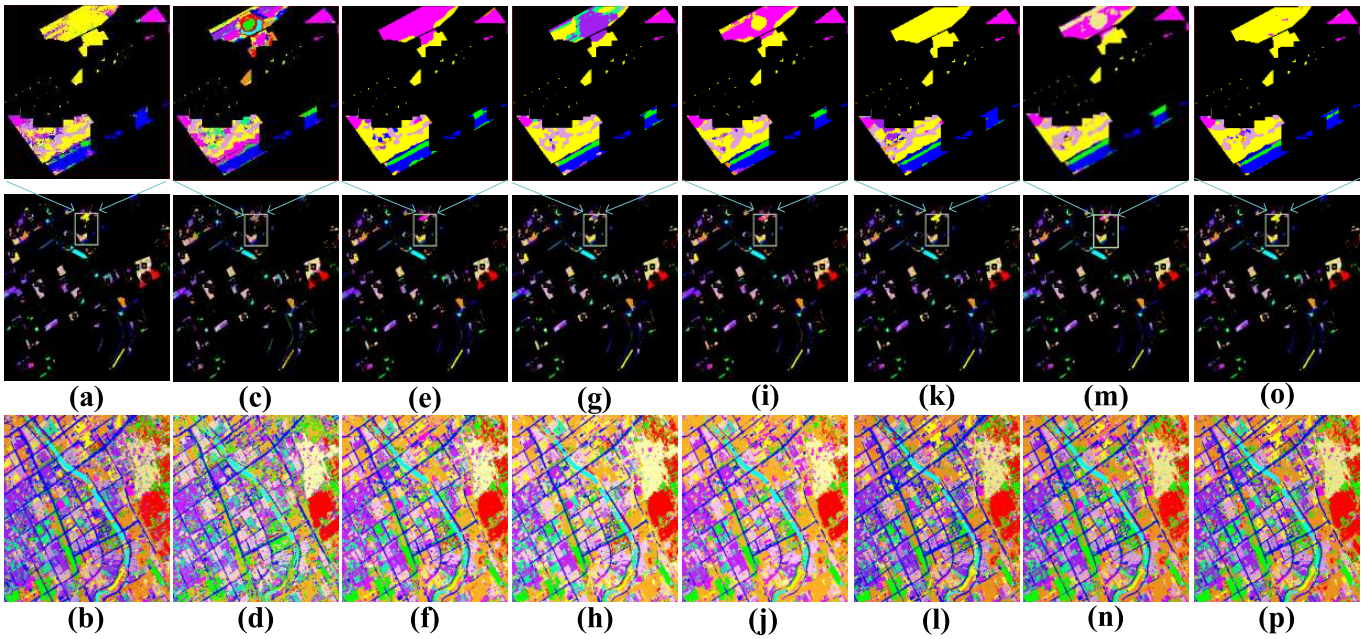


Fig. 12. Maps of classifications generated by various methods for the Hohhot image are presented. For each comparison method, two distinct classification maps are available: one that includes the ground truth map and another that displays the complete classification map. (a) and (b) MFT [50]. (c) and (d) Correspond to the GCFNet [20]. (e) and (f) Illustrate the CRHFF [51]. (g) and (h) Associated with the AM^{3}Net [52]. (i) and (j) Depict the CALC [24]. (k) and (l) Showcase the ISSP-Net [23]. (m) and (n) CMS^{2}I-Mamba [43]. (o) and (p) HCMA-Net.

TABLE V
QUANTITATIVE RESULTS OF THE HOHHOT DATASET

Models	MFT [50]	GCFNet [20]	CRHFF [51]	AM^{3}Net [52]	CALC [24]	ISSP-Net [23]	CMS^{2}I-Mamba [43]	HCMA-Net
c_1: Water	96.18	86.70	95.80	92.02	**96.91**	92.45	96.13	91.99
c_2: Road	76.46	62.20	81.48	76.39	70.45	82.82	82.39	**86.79**
c_3: Bareland	81.99	68.84	88.48	84.44	**95.02**	84.01	89.06	85.90
c_4: Vegetation	69.03	76.07	87.85	84.38	82.52	79.24	**88.28**	85.63
c_5: Intensive Building	54.23	58.99	59.57	56.88	**65.52**	63.79	52.35	53.11
c_6: Sparse Building	60.83	61.91	60.56	65.17	70.69	64.18	**74.02**	68.35
c_7: Suburb Building	67.83	70.10	69.98	**86.66**	67.15	71.92	75.28	77.19
c_8: Farmland	91.36	89.51	93.53	92.59	92.84	95.12	89.93	**95.86**
c_9: White Building	58.79	54.46	55.16	55.63	50.88	62.65	48.31	**71.66**
c_{10}: Red Building	82.46	32.37	78.50	74.00	67.87	72.32	81.55	**83.91**
c_{11}: Blue Building	75.04	32.03	77.11	67.77	72.91	**78.13**	74.59	74.85
OA(%)↑	69.33	63.77	72.21	72.31	72.50	73.77	72.78	**75.26**
AA(%)↑	74.02	63.01	77.09	75.91	75.70	76.97	77.45	**79.57**
Kappa(%)↑	65.47	58.99	68.82	68.95	69.02	70.39	69.42	**72.15**
FLOPs(G)↓	14.68	5.01	**2.44**	43.08	4.35	337.85	18.70	35.04
Times(s)↓	**15.62**	265.81	150.86	81.62	25.25	145.18	66.80	238.74
Parameters↓	212876	3878187	4906823	2777838	**109210**	12082537	1931527	14336923

c_9 (White Building, 71.66%), and c_{10} (Red Building, 83.91%). In addition, since the HCMA-Net has no obvious weak points in the classification results under each category, it achieves the best results in OA (75.26%), AA (79.57%), and Kappa (72.15%). Because the parameters of MFT and CALC are relatively small, their reasoning time is shorter than other methods, while HCMA-Net uses a more multihead attention mechanism, which makes the parameters larger and also greatly increases the reasoning time. However, visual analysis reveals some confusion between White Buildings (c_9) and Blue Buildings (c_{11}). As seen in the zoomed-in regions of classification maps, this is primarily due to spectral similarity caused by lighting conditions and material aging. Furthermore, mixed pixel effects at building edges make it difficult for the model to extract pure spectral features, rendering them susceptible to interference from adjacent blue roof features. Additionally, the performance on c_7 (Suburb Building) also shows room for improvement due to high intraclass variance in size and spatial arrangement.

2) Comparison Results on the Nanjing Image: As shown in Table VI and Fig. 13, the HCMA-Net maintains the highest classification accuracy in the OA (80.38%) and Kappa (77.11%) evaluation indicators and has the highest category accuracy in c_1 (Water, 95.67%), c_4 (Vegetation, 91.14%) and c_7 (Intensive Building, 75.73%) on Nanjing dataset. The OA of HCMA-Net is better than that of models CMS^{2}I-Mamba, ISSP-Net, CALC, AM^{3}Net, CRHFF, which are improved by 0.31%, 5.37%, 1.00%, 10.16%, and 4.86%, respectively. The spectral–spatial mutual-guided module of AM^{3}Net allows it to perform better in c_9 (Sparse Building, 80.99%) and c_{10}

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TABLE VI
QUANTITATIVE RESULTS OF THE NANJING DATASET

Models	MFT [50]	GCFNet [20]	CRHFF [51]	AM^{3}Net [52]	CALC [24]	ISSP-Net [23]	CMS^{2}I-Mamba [43]	HCMA-Net
c_1:Water	94.23	93.45	94.86	88.44	93.80	94.24	95.27	**95.67**
c_2:Road1	64.21	37.36	72.96	40.62	**74.22**	59.47	70.11	65.21
c_3:Bareland	91.84	84.98	94.24	54.10	94.18	**94.50**	92.72	94.14
c_4:Vegetation	76.58	78.65	84.85	71.55	83.72	80.80	88.25	**91.14**
c_5:Red Building	75.54	24.98	77.61	67.85	**88.61**	78.04	88.04	82.43
c_6:Blue Building	91.60	44.95	95.59	93.66	97.71	95.33	**97.89**	96.84
c_7:Intensive Building	64.48	64.42	59.08	74.96	69.51	69.34	70.90	**75.73**
c_8:White Building	75.73	64.74	81.76	90.98	85.08	58.44	**91.88**	65.07
c_9:Sparse Building	38.92	45.62	53.61	**80.99**	40.59	52.80	47.07	56.01
c_{10}:Road2	34.21	13.58	43.07	**91.86**	38.63	33.90	45.84	53.93
c_{11}:Lowvegetation	63.27	47.84	62.74	43.96	**70.92**	59.09	55.15	65.11
OA(%)↑	73.35	61.65	75.52	70.22	79.38	75.01	80.07	**80.38**
AA(%)↑	70.06	54.60	74.58	72.63	76.09	70.54	**76.65**	76.48
Kappa(%)↑	69.11	55.01	71.77	65.02	76.03	71.03	76.76	**77.11**
Times(s)↓	28.57	198.96	206.27	140.97	48.23	286.71	70.93	254.11

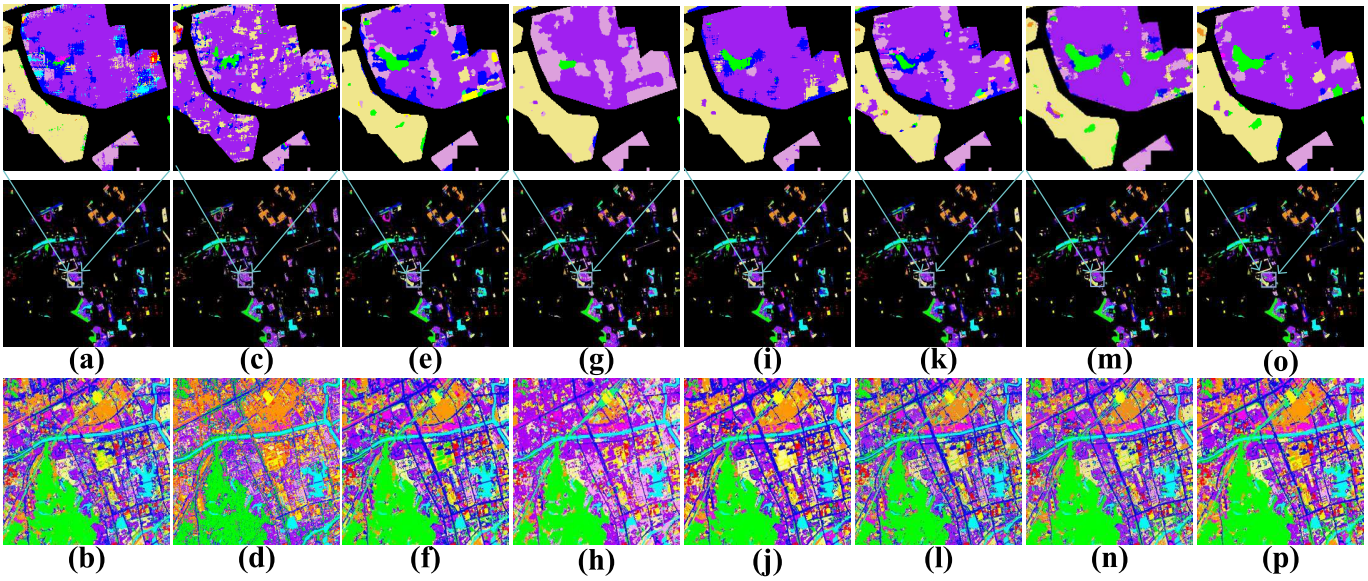


Fig. 13. Maps of classifications generated by various methods for the Nanjing image are presented. For each comparison method, two distinct classification maps are available: one that includes the ground truth map and another that displays the complete classification map. (a) and (b) MFT [50]. (c) and (d) Correspond to the GCFNet [20]. (e) and (f) Illustrate the CRHFF [51]. (g) and (h) Associated with the AM^{3}Net [52]. (i) and (j) Depict the CALC [24]. (k) and (l) Showcase the ISSP-Net [23]. (m) and (n) CMS^{2}I-Mamba [43]. (o) and (p) HCMA-Net.

(Road2, 91.86%) categories with rich semantic information. The multilevel feature fusion classification (MFFC) subnetwork of CALC makes c_2 (Road1, 74.22%), c_5 (Red Building, 88.61%), and c_{11} (Lowvegetation, 70.92%) categories with rich features easier to identify. The pixel-guided spatial enhancement module (PGSE-Module) of ISSP-Net enhances the information representation of spatial location, which in turn enhances recognition in the c_3 (Bareland) category, with a recognition result of 94.50%. Cross-modal feature alignment and semantic consistency ensure that CMS^{2}I-Mamba achieves the highest classification accuracy in the c_6 (Blue Building, 97.89%) and c_8 (White Building, 91.88%) categories, while also achieving the highest (AA, 76.65%) among all algorithms. While HCMA-Net achieves superior performance in categories like Water and Vegetation, maintaining the highest OA and Kappa, it encounters challenges in distinguishing spatially similar categories. Specifically, visual inspection indicates misclassification between Intensive (c_7) and Sparse Buildings (c_9), particularly in transition zones of building density. The patch-based input limits the model's perception of macroscopic spatial density, causing edge samples of intensive areas containing background information to be misidentified as sparse buildings. Similarly, its performance on the c_2 (Road1) category is slightly lower, likely attributable to significant spectral–spatial confusion between roads, shadows, and dark impervious surfaces in high-density urban environments.

3) Comparison Results on the Xi'an Image: As shown in Table VII and Fig. 14, the HCMA-Net relies on excellent feature interaction capabilities, enabling more than one-third of the categories to achieve optimal classification accuracy. At the same time, our network has the best results on OA (83.03%), AA (77.68%), and Kappa (80.64%). HCMA-Net achieves an OA of 83.03%, surpassing CMS^{2}I-Mamba (80.20%), ISSP-Net (79.58%), CALC (78.24%), AM^{3}Net (69.58%), and CRHFF (79.42%) by 2.83%, 3.45%, 5.59%, 13.45%, and 3.61%, respectively. The AM^{3}Net ability to capture spatial locations

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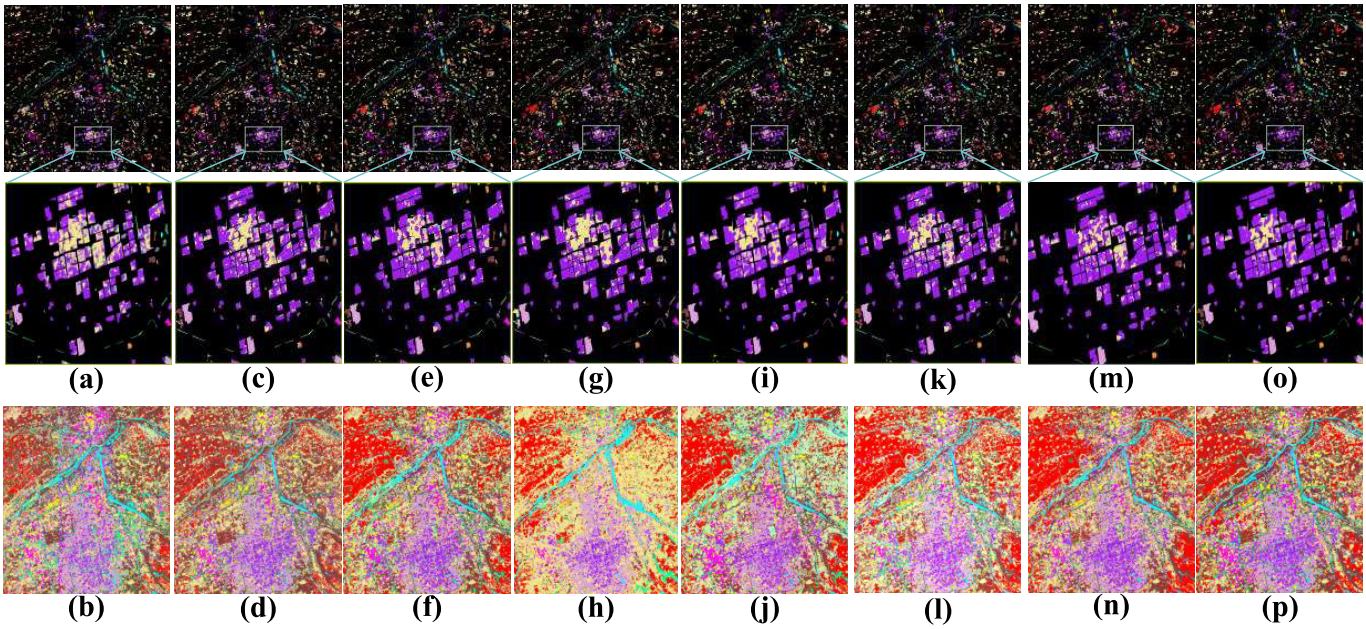


Fig. 14. Maps of classifications generated by various methods for the Xi'an image are presented. For each comparison method, two distinct classification maps are available: one that includes the ground truth map and another that displays the complete classification map. (a) and (b) MFT [50]. (c) and (d) Correspond to the GCFNet [20]. (e) and (f) Illustrate the CRHFF [51]. (g) and (h) Associated with the AM^{3}Net [52]. (i) and (j) Depict the CALC [24]. (k) and (l) Showcase the ISSP-Net [23]. (m) and (n) CMS^{2}I-Mamba [43]. (o) and (p) HCMA-Net.

TABLE VII
QUANTITATIVE RESULTS OF THE XI'AN DATASET

Models	MFT [50]	GCFNet [20]	CRHFF [51]	AM^{3}Net [52]	CALC [24]	ISSP-Net [23]	CMS^{2}I-Mamba [43]	HCMA-Net
c_1:Water	97.35	95.89	95.89	95.61	97.59	96.75	96.14	**98.07**
c_2:Road1	16.82	17.46	18.35	1.72	12.71	26.29	28.21	**46.88**
c_3:Road2	50.69	55.38	62.01	11.61	42.79	68.75	54.50	**70.70**
c_4:Bareland	64.58	66.46	66.45	67.74	39.02	72.70	68.46	**78.90**
c_5:Lowvegetation	46.68	36.06	55.51	18.81	**74.46**	54.81	51.11	34.99
c_6:Farmland	51.69	64.31	69.74	76.35	65.03	71.42	73.85	**80.43**
c_7:Tree	91.78	89.12	84.84	37.73	80.48	86.47	89.48	**92.60**
c_8:Intensive Building	62.85	72.65	77.59	72.44	70.91	74.57	**81.19**	78.43
c_9:Sparse Building	75.73	76.40	77.18	80.85	**82.01**	77.17	76.64	80.73
c_{10}:Suburb Building	91.90	97.81	95.51	**99.81**	96.93	95.47	95.65	98.47
c_{11}:Workshop Building	94.52	90.71	95.39	75.48	**96.74**	92.03	93.03	96.26
c_{12}:White Building	72.87	66.36	75.33	45.00	74.14	67.29	**77.63**	75.74
OA(%)↑	74.65	76.86	79.42	69.58	78.24	79.58	80.20	**83.03**
AA(%)↑	68.12	69.05	72.81	56.93	69.40	73.64	73.83	**77.68**
Kappa(%)↑	71.08	73.48	76.44	64.18	75.11	76.66	77.32	**80.64**
Times(s)↓	**140.23**	2474.43	1159.44	1041.57	199.83	872.85	602.27	1458.46

drives extremely good categorization in categories c_{10} (suburb building, 99.81%). The CALC has the highest accuracy on c_5 (Lowvegetation, 74.46%), c_9 (Sparse Building, 82.01%), and c_{11} (Workshop Building, 96.74%). This is because the CALC fully exploits and mines high-level semantic information and complementary information, while also increasing adversarial training. It effectively preserves detailed information in PAN and MS images. The CMS^{2}I-Mamba achieved the best classification results in both c_8 (Intensive Building, 81.19%) and c_{12} (White Building, 77.63%) categories due to the use of channel interaction mechanism to ensure feature alignment of heterogeneous modalities. The HCMA-Net relies on excellent feature interaction, enabling more than one-third of the categories to achieve optimal classification accuracy, with the best OA and Kappa. However, visual analysis suggests that confusion persists between Intensive (c_8) and Sparse Buildings (c_9). Atypical spatial arrangements and uneven spacing in some intensive zones lead to texture overlap with sparse buildings. The model's reliance on local patch features occasionally overlooks the broader semantic context required to distinguish these spatially dense but spectrally similar categories. Additionally, the results for c_5 (Lowvegetation) indicate a challenge stemming from high spectral similarity with other vegetation-based classes and boundary ambiguity.

4) Comparison Results on the Houston2013 Image: As shown in Fig. 15 and Table VIII, the HCMA-Net has the highest result on c_2 (Stressed Grass, 100%), c_7 (Residential, 94.68%) and c_{15} (Running Track, 100%). This proves that the overall effectiveness of our method is stronger than other methods. The OA of the HCMA-Net outperforms the mainstreamed models CMS^{2}I-Mamba, ISSP-Net, CALC, AM^{3}Net,

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TABLE VIII
QUANTITATIVE RESULTS OF THE HOUSTON2013 DATASET

Models	MFT [50]	GCFNet [20]	CRHFF [51]	AM^{3}Net [52]	CALC [24]	ISSP-Net [23]	CMS^{2}I-Mamba [43]	HCMA-Net
c_1: Health Grass (198/1053)	82.21	80.53	81.58	84.04	80.53	**85.00**	82.81	82.62
c_2: Stressed Grass (190/1064)	84.99	95.39	95.49	99.06	95.02	94.17	99.44	**100.00**
c_3: Syntheic Grass (192/505)	99.08	92.28	90.69	**99.21**	93.86	96.83	94.65	99.01
c_4: Trees (188/1056)	96.28	95.64	96.50	**99.34**	98.67	94.51	98.96	95.08
c_5: Soil(186/1056)	99.72	99.72	99.62	99.05	99.15	**100.00**	**100.00**	98.01
c_6: Water(182/143)	94.87	**100.00**	99.30	100.00	97.20	94.41	90.21	95.80
c_7: Residential(192/1072)	83.46	85.54	80.60	90.76	82.37	89.27	87.41	**94.68**
c_8: Commercial(191/1053)	81.70	**94.59**	92.69	87.75	87.37	92.31	97.25	90.12
c_9: Road(193/1059)	90.71	84.61	82.81	95.28	91.60	**94.15**	88.67	93.20
c_{10}: Highway(191/1036)	54.12	73.36	74.13	69.79	82.63	**93.63**	76.25	93.24
c_{11}: Railway(181/1054)	98.48	98.67	**100.00**	78.46	97.34	93.93	95.16	93.17
c_{12}: Parking lot 1(192/1041)	**93.95**	89.24	86.07	82.32	87.70	77.43	72.43	87.51
c_{13}: Parking lot 2(184/285)	86.43	72.98	74.74	90.18	88.77	85.61	**91.23**	85.26
c_{14}: Tennis court(181/247)	99.06	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.60
c_{15}: Running Track(187/473)	93.09	91.12	91.33	99.79	97.25	**100.00**	**100.00**	**100.00**
OA(%)↑	87.72	89.85	89.14	89.97	90.92	92.08	90.72	**93.32**
AA(%)↑	89.21	90.24	89.70	91.66	91.96	92.75	91.63	**93.82**
Kappa(%)↑	86.67	88.97	88.21	89.05	90.13	91.40	89.92	**92.74**
Time(s)↓	37.68	22.58	45.21	52.14	**22.17**	63.11	98.52	66.21

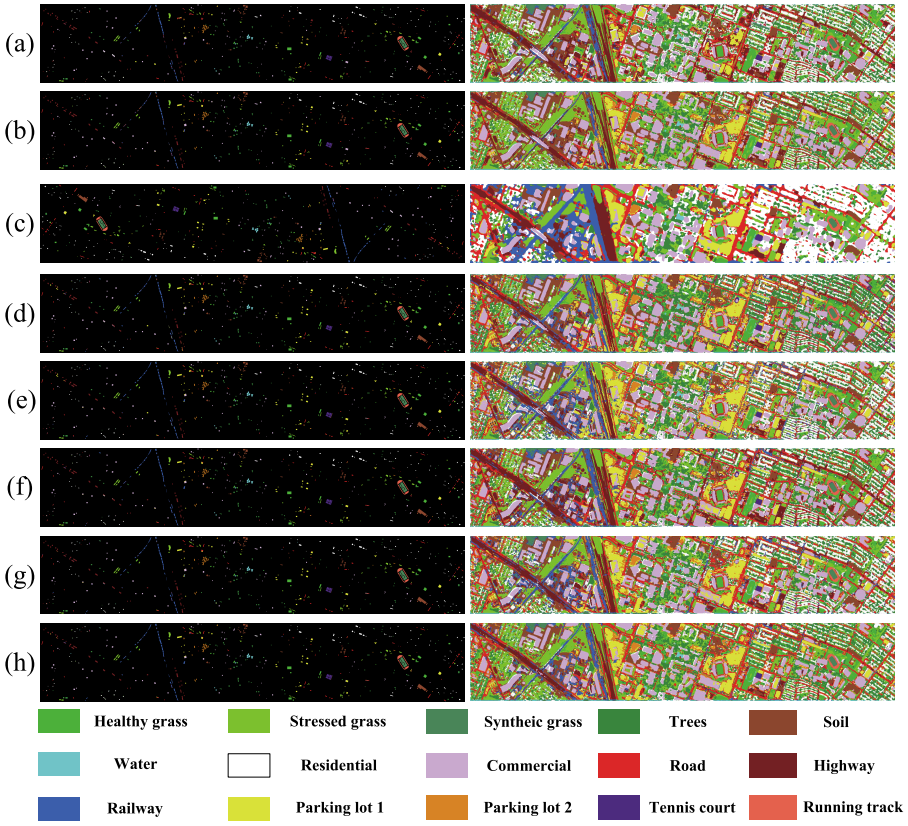


Fig. 15. Maps of classifications generated by various methods for the Houston2013 images are presented. Method (a)–(h) corresponds to the approaches listed in Table VIII. The first column highlights the label-specific predictions. The second column presents the overall prediction maps.

CRHFF, GCFNet and MFT by 2.6%, 1.24%, 2.4%, 3.35%, 4.18%, 3.47%, 5.6%, respectively. However, HCMA-Net performs poorly in the classification of the c_4 (Trees) and c_5 (Soil) categories. Visual analysis confirms that these errors occur largely in mixed scenes of sparse vegetation and soil. The multiscale fusion may introduce information redundancy or noise propagation in these areas. Specifically, low-level textural details of soil may interfere with high-level semantic identification of trees, resulting in feature vectors from different categories overlapping in the representation space and becoming difficult to distinguish in transitional boundaries.

F. Quantitative and Qualitative Experiment on Feature Space Alignment

To further investigate the reasons behind the outstanding performance of HCMA-Net in classification tasks, we conducted a study on the feature spaces occupied by PAN and MS, using the Hohhot dataset as an example. First, we investigate and demonstrate whether different modal samples of the same category cluster to measure interclass separability. It is expected that different categories should maintain clear separation, while different modalities within the same category should cluster together. Subsequently, we observed the overall

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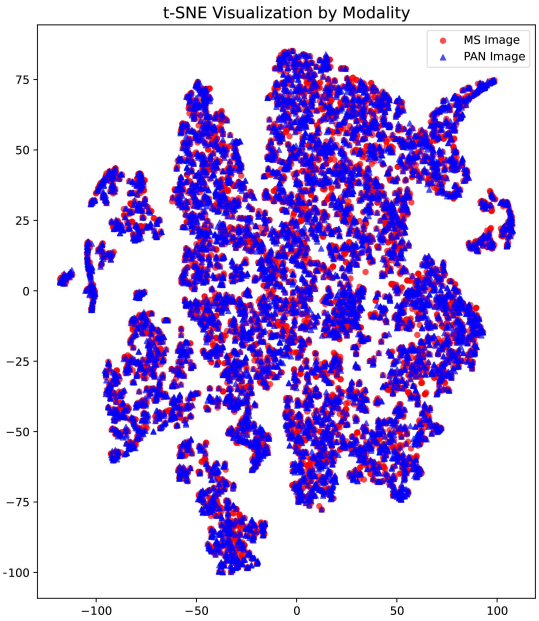


Fig. 16. t-SNE visualization of PAN-MS distribution with HCMA-Net on Hohhot dataset. For the convenience of display, we selected 4000 samples from 274 045 test samples for visualization.

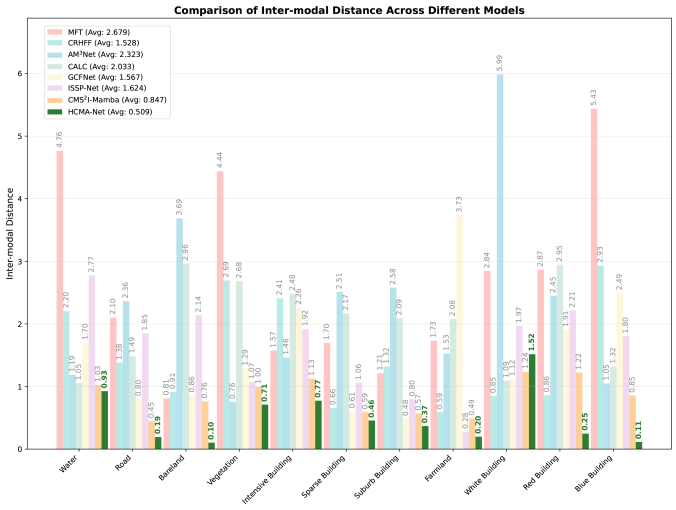


Fig. 17. Comparison of intermodal distance across eight models on the Hohhot dataset for 274 045 test samples. The ideal situation is that the distance between different modalities of the same class is 0.

distribution patterns of the two modalities to reflect the effect of modal fusion. Complete separation indicates significant differences in modal features, while overlap indicates good feature alignment. Finally, to further highlight the advantages of HCMA-Net in feature space alignment, we calculated and displayed the specific intermodal distances for all comparative algorithms in each category. A smaller distance indicates that the two modal features of that category are more similar to each other.

1) Qualitative Analysis: Fig. 16 shows that the spatial distribution of spatial and spectral modalities overlaps more, and the boundaries between modalities become blurred (Compared to Fig. 1), indicating a decrease in modal differences in the feature space. This increase in overlap indicates that the model has better learned a unified representation across modalities.

2) Quantitative Analysis: As shown in Fig. 17, first, HCMA-Net exhibits the lowest inter-modal distance among all categories, with an average distance of only 0.509, significantly better than other comparative models, demonstrating its effectiveness in reducing semantic differences between modalities. A lower intermodal distance means that the model can better align and fuse feature representations from different modalities, thereby improving the effectiveness of multimodal learning.

Meanwhile, there are significant differences in the difficulty of modal matching between various categories. For example, in categories such as Sparse Building and Blue Building, the modal distances of all models are relatively low, indicating that the cross-modal features of these categories are easier to align. In categories such as Water, Vegetation, and White Building, most models exhibit higher intermodal distances, indicating that modal space alignment in these categories is more challenging.

In conclusion, HCMA-Net not only demonstrates superior average performance but also maintains a high degree of stability across all categories. Its modal distance values are consistently low, signifying its ability to reliably achieve effective modal matching across various categories.

A comprehensive analysis of the experimental results shows that HCMA-Net achieves superior classification accuracy, particularly in overcoming challenges associated with multimodal feature alignment and integration.

V. Conclusion

HCMA-Net, a novel HCMA-Net, tackles the representation gap in multimodal remote sensing image classification. Its design includes two key parts: the HFAE-Module and the CMIFE-Module. HCMA-Net reduces feature bias by connecting detail-semantic information and balances features within and between modalities through cross-modal feature interaction, thereby improving classification performance. Tests on multiple datasets show it outperforms existing methods. However, our method relies on low-level visual features alone, without leveraging external semantic information from class labels or text prompts, which could serve as a powerful bridge to guide cross-modal feature alignment and fusion. We plan to explore a vision-language fusion framework to effectively integrate text-based cues into our network, aiming for synergistic learning between visual features and linguistic information to further enhance classification performance.

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